Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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Executive Summary

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Time-Series Forecasting

Time-series forecasting underpins operational decision-making wherever a system's future state must be anticipated from its past, from grid load to industrial condition monitoring: projecting historical sequences into future horizons is what allows a control policy to be proactive rather than reactive[1].

During the preparation of this work, the authors used AI-based tools, including GPT-5.6 Sol, Gemini 3.7 Flash, Sonnet 5, and Opus 5, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Evolution of Forecasting Methodologies

Classical time-series analysis relies on statistical models such as AutoRegressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS)[1]. These are interpretable and robust on stable low-dimensional data, but they assume linearity and stationarity and are fitted to one series at a time.

Deep learning relaxed those assumptions: recurrent networks and their gated variants[2] model non-linear dynamics through a hidden state, and training jointly across many related series established the global forecasting paradigm[3]. Transformers then replaced the recurrent bottleneck with self-attention[4], computing dependencies between distant time steps directly. Two costs follow for sampled signals: attention scales quadratically with sequence length, and a single time step is a weak unit of meaning. Patching addresses both, segmenting the series into contiguous sub-sequences and embedding each as one token, which shortens the sequence and gives every token a locally semantic support[5].

Time-series foundation models sit at the end of this line. Chronos treats time-series values as discrete tokens obtained by scaling and quantisation[6]; its successor Chronos-Bolt replaces quantisation with the patching scheme above and a direct multi-quantile head[7].[1] Patching is therefore not an incidental implementation detail, but the design choice that makes this class of models practical at scale.

Classic Aliasing

Whatever the architecture downstream, predictive fidelity is bounded by the digital representation of the signal. The governing limit is the Nyquist-Shannon sampling theorem[9]: a band-limited continuous-time signal is perfectly reconstructible only if sampled uniformly at a rate f_s strictly greater than twice its highest frequency component $f_{\max}$,

$$f_s > 2f_{\max}. \tag{1}$$

The threshold $f_N = f_s/2$ is the Nyquist frequency. When the criterion is violated, through an insufficient sampling rate or a missing anti-aliasing filter, components above f_N fold back into the lower spectrum and become indistinguishable from genuine low-frequency content[10]. This is *aliasing*, and it is irreversible: the information is destroyed at acquisition, not at inference. Respecting Equation 1 is consequently taken as sufficient for a discrete sequence to represent its source faithfully. This report asks whether that sufficiency survives patch-based tokenization.

Application Domain and Scope

The reference setting is a designed scenario: a photovoltaic array supplying an internal load, a local storage battery and the national grid, with a control agent performing load balancing, battery dispatch, state monitoring and anomaly detection on short-horizon forecasts. Measured photovoltaic and weather telemetry are taken from the SolarTech Lab dataset[11]; the concepts, states and relations of the scenario are formalised as an OWL2 ontology in Phase locking and Aliasing.

The target architecture is Chronos-Bolt-Tiny, with default patch size $P = 16$, stride $S = 16$, context window $T = 2048$, four encoder and four decoder blocks, and a direct quantile projection head[12]. The objective is to test the assumption that such a model preserves the frequency content of Nyquist-compliant signals, and to characterise how f_s, P and S interact in determining when it does not[13].

[1] The vendor reports up to a 250× inference speed-up over the original Chronos[8]; no peer-reviewed description of Chronos-Bolt has been published.

Patch-Based Tokenization: a Formal Description

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$.

The k-th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix}. \tag{2}$$

Using element-wise indexing notation,

$$\mathbf{t}_k[m] = x[kS+m], \qquad m \in \{0,\ldots,P-1\}. \tag{3}$$

Before examining the degeneracies of this operator it is worth fixing what it does not do. Whenever $S \leq P$, every sample of the input appears in at least one token and can be read back from it: taking $k = \lfloor n/S \rfloor$ and $m = n \bmod S \leq S-1 \leq P-1$ in Equation 3 gives

$$x[n] = \mathbf{t}_{\lfloor n/S \rfloor}[n \bmod S], \qquad S \leq P. \tag{4}$$

Complete raw patching with $S \leq P$ is therefore injective: the token sequence determines the signal, and no frequency below Nyquist is destroyed by tokenization itself. Everything that follows concerns redundancy in the token sequence rather than loss of information, the single exception being the $S > P$ regime of Case 3 below, where the operator does discard samples.

Phase locking and Aliasing

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component and model it as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{5}$$

where f denotes the signal frequency satisfying the Nyquist criterion $f < \frac{f_s}{2}$, and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S. Evaluating a stride translation yields

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \tag{6}$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of 2π-periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \qquad c \in \mathbb{N}^+. \tag{7}$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \tag{8}$$

The same derivation can be applied taking into account translational shifts equal to the patch size P, yielding a similar phase-locking condition

$$2\pi \frac{fP}{f_s} = 2\pi k \Rightarrow x[n+P] = x[n], \qquad k \in \mathbb{N}^+. \tag{9}$$

Solving Equation 7 and Equation 9 for f yields the phase-locking frequency class $\mathcal{F}_{\text{lock}}$, which is defined as the set of frequencies producing self-repetition over one stride or over one patch.

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f = c\frac{f_s}{S} \vee f = k\frac{f_s}{P},\ c, k \in \mathbb{N}^+,\ f < \frac{f_s}{2} \right\}. \tag{10}$$

It is convenient to express both conditions in units that do not depend on the sampling rate. Introducing the cycles-per-stride and cycles-per-patch ratios

$$\text{cps} = \frac{fS}{f_s}, \qquad \text{cpp} = \frac{fP}{f_s}, \tag{11}$$

Equation 7 and Equation 9 read simply cps $\in \mathbb{N}^+$ and cpp $\in \mathbb{N}^+$: $\mathcal{F}_{\text{lock}}$ is the set of frequencies completing a whole number of cycles within one stride or within one patch. This normalisation makes the two branches directly comparable across geometries and across sampling rates.

Neither branch is restricted to sinusoidal signals. Both conditions generalise to an arbitrary discrete-time signal invariant under a translation equal to the stride or to the patch,

$$x[n+S] = x[n] \qquad \text{or} \qquad x[n+P] = x[n], \qquad \forall n, \tag{12}$$

but only the first bears on consecutive patches. Tokens begin at multiples of the stride, so the step from token k to token $k+j$ is a shift of jS samples. Substituting Equation 3 and reducing by the stride condition at $j = 1$ gives

$$\mathbf{t}_{k+1}[m] = x[(k+1)S + m] = x[kS + m + S] = x[kS + m] = \mathbf{t}_k[m], \tag{13}$$

and since this holds for every component,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \tag{14}$$

The patch condition reduces the same shift only when P divides jS. At $j = 1$ that requires $P \mid S$, which under $S \leq P$ occurs only at $S = P$, so consecutive tokens are left distinct. It does hold at the smallest j for which $P \mid jS$, namely $j = P/\gcd(P, S)$, giving

$$\mathbf{t}_{k+j}[m] = x[(k+j)S + m] = x[kS + m + jS] = x[kS + m] = \mathbf{t}_k[m], \qquad j = \frac{P}{\gcd(P, S)}, \tag{15}$$

and therefore

$$\mathbf{t}_{k+j} = \mathbf{t}_k. \tag{16}$$

Patch-periodicity thus makes tokens equal as well, but at a spacing of j rather than between neighbours. The two conditions coincide on consecutive patches only in the non-overlapping geometry, where $j = 1$.

The derivations above show that some translations leave a signal unchanged; by Equation 4 this repetition is not, by itself, evidence that two distinct signals become indistinguishable to the model. Structural aliasing is therefore treated as an emergent, falsifiable property of the trained representations[13]: it is diagnosed by readability rather than by a distance between raw tokens, and it is present when the signal's frequency can no longer be recovered from the model's internal state at a locked frequency as well as it can at neighbouring, non-locked ones. $\mathcal{F}_{\text{lock}}$ is accordingly a set of candidate frequencies, derived from the geometry alone; whether the candidates are realised is an empirical question.

Analysis of Structural Redundancies and Invariances

Equation 10 is the union of two distinct families,

$$\mathcal{F}_S = \left\{ c\frac{f_s}{S} \right\}_{c\in\mathbb{N}^+}, \qquad \mathcal{F}_P = \left\{ k\frac{f_s}{P} \right\}_{k\in\mathbb{N}^+}, \qquad \mathcal{F}_{\text{lock}} = (\mathcal{F}_S \cup \mathcal{F}_P) \cap \left[0, \tfrac{f_s}{2}\right), \tag{17}$$

which coincide if and only if $S = P$, and which act on the token sequence in different ways. A frequency in $\mathcal{F}_S$ satisfies the stride condition, so by Equation 13 every token equals its predecessor: the sequence becomes one window repeated, and the number of distinct tokens drops to one however long the context. A frequency in $\mathcal{F}_P$ has a period $T_0 = f_s/f$ dividing P, so it satisfies the patch condition and by Equation 16 the sequence instead repeats every $j = P/\gcd(P, S)$ tokens, cycling through at most that many distinct vectors. Each token still holds a whole number of cycles, but neighbours read that waveform from different starting points and coincide only when $j = 1$. The two branches therefore differ in what they repeat rather than in whether they repeat: the stride branch acts between neighbouring tokens, the patch branch on the sequence as a whole.

Neither removes information. For $S \le P$ both are regimes of Equation 4, so the signal remains recoverable and two inputs with the same token sequence are the same signal on the observed support. Both repetitions belong to the raw tokens, not to the learned representation; treating either as an invariance of the model would require showing that the patch embedding is itself shift-equivariant, which is neither established here nor assumed.

Case 1: Equal Patch Size and Stride ($P = S$) When patches are contiguous and non-overlapping the two branches coincide, $\mathcal{F}_S = \mathcal{F}_P$, and $j = P/\gcd(P, P) = 1$, so Equation 16 collapses onto Equation 14. This is the only geometry in which both conditions act on consecutive patches, and every locked frequency then gives

$$\mathbf{t}_k = \mathbf{t}_0, \qquad \forall k. \tag{18}$$

This is the most redundant geometry of the family: the only one in which every locked frequency reduces the token sequence to a single repeated vector, leaving no residual structure across tokens. It is also the default configuration of the target architecture, which operates at $P = S = 16$.

Case 2: Overlapping Windows ($S < P$) When the stride is smaller than the patch length the two branches separate. Since $f_s/S > f_s/P$, the stride comb is the sparser of the two: strictly below Nyquist it contributes $\lceil S/2 \rceil - 1$ members against the $\lceil P/2 \rceil - 1$ of the patch comb, and whenever S divides P the stride comb is a subset of the patch comb. Most locked frequencies in this regime therefore belong to $\mathcal{F}_P \setminus \mathcal{F}_S$, where consecutive tokens remain distinct and the sequence merely repeats with period $j > 1$.

Overlap does not weaken the exact condition. The derivation of Equation 13 makes no use of the relation between P and S, so at $f \in \mathcal{F}_S$ consecutive tokens are still exactly equal. What overlap changes is the behaviour around a lock: adjacent windows share $P - S$ samples by construction, so two consecutive tokens are constrained to agree on their common support regardless of frequency, and the transition into repetition is correspondingly smoother. Reducing the stride at fixed patch size therefore raises the lowest member of $\mathcal{F}_S$, thins the stride comb below Nyquist, and softens the local shape of the effect without removing it.

Case 3: Disjoint Windows ($S > P$) When the stride exceeds the patch length the windows leave $S - P$ unobserved samples between consecutive patches, and Equation 4 fails: $\mathcal{T}_{P,S}$ is non-injective for every f, since signals differing only on the gaps share a token sequence. The loss is one of discarded samples rather than of tokenization geometry, so these configurations are excluded by construction from the experimental design.

What the geometry does and does not establish The three cases can be summarised in four statements, of which only the last is an empirical claim. $\mathcal{F}_S$ collects the stride-lock candidates, at which consecutive raw tokens repeat. $\mathcal{F}_P$ collects the frequencies whose period divides the patch, at which the sequence repeats every $P/\gcd(P, S)$ tokens while neighbours stay distinct unless $S = P$. Their union $\mathcal{F}_{\text{lock}}$ is therefore a statement about candidate geometry, not a proof of aliasing: for $S \leq P$ the raw token sequence remains a lossless encoding of the input at every one of these frequencies. Structural aliasing is the separate, representation-level proposition that a model trained on such sequences nevertheless fails to make those frequencies readable from its internal state. It is a property of the learned representation rather than of the operator, and one that can only be established empirically. The hypotheses that test it are formulated in Bayesian Analysis.

Ontology Domain Description

The ontology reproduced in Appendix A closes the loop from the signal to the decision. It represents the photovoltaic micro-grid, its telemetry and a single control agent, and it makes the tokenisation geometry a first-class object rather than a property of the model: a `pv:TimeSeriesSignal` has a `pv:PatchedRepresentation` carrying `pv:patchSize`, `pv:stride` and a derived `pv:overlapRatio`, and one signal may hold several, one per (P, S) under evaluation. Since the lock conditions are arithmetic on those same fields, the set of candidates is derivable from the knowledge base rather than hard-coded beside it, which is what lets an agent recompute it when a configuration or a sampling interval changes.

The point of contact with the Bayesian analysis is the following. An `pv:ObservabilityAssessment` does not derive a state from the geometry alone. It records the estimand, its effect size, its credible interval and its posterior probability, and assigns StructurallyAliased only to a frequency that is a lock candidate and also carries confirmed evidence of loss, $X = Z \wedge Y$ in the notation of Appendix A. Geometry alone yields a candidate, never a verdict. Because the confirmation threshold exceeds one half, confirmed loss and confirmed preservation cannot both hold. A frequency for which preservation is positively confirmed $(\neg X \wedge R)$ is assigned Observable; everything else is assigned PartiallyObservable: an unfinished test, a credible interval crossing the threshold, or conflicting metrics. That is the inconclusive verdict of Table 3 carried into the semantics, and its consequence is operational rather than documentary, because the assessment selects the control mode. Confirmed aliasing forces the safe mode and suspends forecast-driven battery decisions; a merely partially observable assessment reduces the battery cap and lowers decision confidence. Uncertainty about the representation is therefore propagated to the actuator instead of being resolved by assumption.

Bayesian Analysis

Bayesian inference is used to test the structural-aliasing hypotheses while making the analysis assumptions and uncertainty explicit: prior assumptions are stated for each estimand, posterior uncertainty is quantified for every estimated effect, and competing hypotheses and the mitigation claim are compared without relying only on point estimates or null-hypothesis significance testing.

Each hypothesis receives its own measurement, its own model and its own estimand. Table 1 lists the models with the value each claim predicts, Table 2 every prior, and Table 3 the rule that turns a posterior into a verdict. All are ordinary regressions; only the quantity on the left and the question the coefficient answers change between them.

Model	**Claim**	**Estimand**	**Predicts**	**Input**
A	$H1$ behavioural	$e^{\bar{\beta}}$: recovery at a lock / at its controls	< 1	`contrasts`
B	$H1$ representational	$e^{\theta_{lock}}$: codelength at a lock / elsewhere	> 1	`mdl_cells`
C	$H2$	σ_ϕ: spread of the deficit across phase	≈ 0	`contrasts`
D1	$H3$ location	θ_S, θ_P: dip depth on each grid, by LOO	both < 0	`collapse`
D2	$H3a$ stride branch	κ_S: measured / stride-predicted spacing	$= 1$	`sites`
D2	$H3b$ patch branch	κ_P: measured / patch-predicted spacing	$= 1$	`sites`
A′	$M1$ mitigation	δ_O: change in the deficit per unit of overlap	< 0	`contrasts`

Table 1: *The models, the quantity whose posterior decides each claim, the value the claim predicts, and the probing table each is fitted to.* $H1$ *is tested twice, on the forecast (A) and on the representation (B), so that a discrepancy between them stays visible.* $H3$ *is split into its two branches, since* $\mathcal{F}_{lock}$ *is a union and a frequency may belong to either.* $M1$ *is not a separate model but the configuration level of A, promoted to a named estimand.*

Each model is built the same way: a measurement, a linear predictor μ_i that adds one offset per group the observation belongs to, and a likelihood. One coefficient is the estimand and the rest keep it unconfounded. Table 2 gives every prior with the role its parameter plays. The range over which the prior scale is varied as a sensitivity check is $\{0.25, 0.5, 1.0\}$; all are centred on no effect, so a posterior that has not moved from its prior is itself a report of an absent effect.

H1

The behavioural measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$, collected in matched triplets: each lock f_k with its two controls, all three sharing the same background realisation and the same phase, which is what makes the contrast paired. The contrast used is

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right], \tag{19}$$

one number per triplet, negative when the lock recovers less than its neighbours. It can be formed only because the injected tone has a known frequency. Model A fits it as

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big). \tag{20}$$

Equation 20 has four components, each motivated separately.

- **Three offsets in μ_i:** β_c per configuration, u_k per lock harmonic and u_b per background draw, so that neither an unusual frequency nor a single corpus can be mistaken for the effect.
- **Student-t_4, not Gaussian.** Where the model rebuilds nothing, recovery is near zero and the logarithm returns very large contrasts; under a Gaussian a small number of them would determine the estimate.
- β_c **drawn around** $\bar{\beta}$, making the geometries instances of one phenomenon rather than unrelated experiments. Two consequences follow. $\bar{\beta}$ exists as a parameter, which $H1$ requires, being a claim about patch-based tokenisation and not about a single checkpoint; and each geometry contributes equal weight, so the largest design cannot dominate the estimate when the number of usable triplets varies several-fold with the in-band lock count.
- **Two configuration-level covariates**, estimable together rather than confounded because the design varies overlap and patch size independently. Overlap enters as a ratio, patch size as $\log P$ because the patch grid has spacing f_s/P and equal steps in $\log P$ are equal ratios of spacing. The tilde denotes centring on the design, $\widetilde{O}_c = (O_c - \bar{O})/0.5$ and $\widetilde{\log P}_c = \log P_c - \overline{\log P}$, which leaves $\bar{\beta}$ the effect at the average configuration rather than an extrapolation to $O = 0$ and $P = 1$.

Model	Parameter	Prior	Role
A, C	$\bar{\beta}$	Student-$t_4(0, 0.5)$	population phase-lock effect, on the log-recovery scale
A	δ_O	Student-$t_4(0, 0.5)$	slope in $\widetilde{O}_c$, the centred overlap; one unit $= 0.5$ of O
A	δ_P	Student-$t_4(0, 0.5)$	slope in $\widetilde{\log P}_c$, the centred log patch size
A, C	$\tau, \sigma_k, \sigma_b, \sigma$	Half-Student-$t_4(0, 0.5)$	configuration, harmonic, background and residual scales
C	σ_ϕ	Half-$\mathcal{N}(0, 0.25)$	spread of the eight per-phase offsets
B	α_0	$\mathcal{N}(\log \bar{y}, 1)$	baseline log codelength
B	θ_{lock}	$\mathcal{N}(0, 0.5)$	log codelength expansion at a lock
B	k	Gamma$(2, 0.1)$	Gamma shape (dispersion)
B	$\sigma_{st}, \sigma_{geo}$	Half-$\mathcal{N}(0, 1)$	spread of the probe-stage and geometry offsets
D1	$\alpha_g, \theta_S, \theta_P$	$\mathcal{N}(0, 1)$	per-geometry off-grid level and the two grid labels
D1	σ	Half-$\mathcal{N}(0, 1)$	residual scale of $\log z_g(f)$
D2	κ_S, κ_P	$\mathcal{N}(0, 1)$	measured spacing per unit of predicted spacing
D2	σ_F	Half-$\mathcal{N}(0, 5)$ Hz	scatter beyond the sweep-resolution floor Δf

Table 2: *Every parameter with its prior and its role. The effect priors are those of Deliverable 1, extended by δ_P, which the design of Table 4 makes estimable.*

Model B poses the same question of the representation, on a frequency-local prequential codelength: at each f_c a probe separates tones at $f_c \pm 1$ Hz from a captured internal state. Ten states are probed: the encoder [REG] token after each encoder block, the decoder state after each decoder block, the pre-projection vector and the quantile head; the decoder-side vectors are named decoder states because Chronos-Bolt carries [REG] in the encoder only. The regression is Gamma with a log link, so $e^{\theta_{lock}}$ is the factor by which the codelength of a locked frequency increases, with probe stage and geometry as zero-mean offsets since codelengths differ several-fold between stages. Deliverable 1's band tasks summarise hundreds of frequencies at once, so the lock indicator has nothing to attach to; they are kept as a cross-check.

H2

$H2$ uses the same triplets with one change. Equation 19 has already averaged over phase, so the phase index is kept instead and the response becomes the per-phase deficit $y_i = \log(R_{\text{ctrl},i} + 0.01) - \log(R_{\text{lock},i} + 0.01)$, which is d with its sign reversed and positive when the lock is worse. Model C is Model A on that response, without the configuration-level covariates and with the phase circle cut into eight slices, each given its own offset u_p:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{21}$$

The estimand σ_ϕ is the spread of those eight offsets, and it answers the question with one number. If the deficit depends on where in its cycle the signal starts, the offsets differ and σ_ϕ is large. If the geometry alone determines it, they are equal and σ_ϕ is near zero, as $H2$ predicts. Slices are used rather than a fitted curve because they assume nothing about the form a dependence might take. Dropping u_p gives a nested null, and comparing the two by leave-one-out cross-validation (LOO) separates the size of the phase effect from the evidence that it exists.

H3

$H3$ was stated in Deliverable 1 as two claims, one per branch. Both are tested on the token collapse $z_g(f)$, the dispersion of the input-patch embeddings across patches, which is zero when consecutive patches are identical. All geometries share one sweep grid: a uniform 1 Hz sweep unioned with the predicted sites of every configuration swept. No geometry is therefore scored only where its own theory predicts a dip, and non-integer sites such as $42.\overline{6}$ Hz for S=12 are covered.

The first question is where the dips are located. Each swept frequency carries two labels, one per branch:

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \text{OnStrideGrid} + \theta_P \cdot \text{OnPatchGrid},\ \sigma)\,. \tag{22}$$

The logarithm makes the labels multiplicative on the per-geometry off-grid level α_g. The two label coefficients θ_S and θ_P are the estimands, and $H3$ predicts both negative, so the two-label fit should win the LOO comparison against each label alone and against neither. The two labels are not competing explanations of the same dips, since a frequency may carry either or both. The two coefficients separate, however, only where exactly one label applies, which for θ_S means the geometries with $S \nmid P$.

The second question is whether the dips move. The fit above remains compatible with dips that merely sit on a fixed grid, so each branch is checked against its own scaling law. Every detected dip is assigned to the branch that predicts it, and sites belonging to both are set aside. The fundamental $\hat{f}^{F}_{1,g}$ of branch F in geometry g is then compared with the spacing that branch predicts, $\Delta^S_g = f_s/S_g$ or $\Delta^P_g = f_s/P_g$:

$$\hat{f}^{F}_{1,g} \sim \mathcal{N}\left(\kappa_F\,\Delta^F_g,\ \sqrt{\sigma_F^2 + \Delta f^2}\right). \tag{23}$$

The slope κ_F is the estimand, measured spacing per unit of predicted spacing, so $\kappa_F = 1$ denotes a branch that tracks its own arithmetic. The scatter σ_F about it is given a floor Δf, the sweep step, since a spacing read off a discrete grid is no more precise than one step; without the floor, sites landing exactly on the prediction drive σ_F to zero and the posterior degenerates.

***H3a*, the stride branch** The prediction is $\kappa_S = 1$: the stride sites track f_s/S one for one, so a site that remains fixed while S changes refutes it. Identification needs a series that holds P fixed and varies S, which Table 4 supplies at P=16, 24 and 32.

***H3b*, the patch branch** The prediction is $\kappa_P = 1$ in the same way. Identification needs a series that holds S fixed and varies P, which Table 4 supplies at S=8, 12 and 16.

Decisions and validation

Every threshold in Table 3 is fixed before a posterior is inspected, and validation comes before interpretation. Models A and C include a per-background random offset u_b with scale σ_b; this lets the posterior separate generator-specific variation from the aliasing effect itself, so that a recovery difference between Light TSMixup and KernelSynth backgrounds is attributed to the generator rather than inflating or deflating the estimate of $\bar{\beta}$. The experimental grid, sampling parameters and generators are described in Signal Generation and Testing.

Notebook workflow

The Bayesian analysis is implemented in one reproducible notebook. Its role is not to introduce a new method after the fact, but to execute the hypotheses, measurements, priors and decision rules defined above in a fixed order. The notebook therefore has a short setup stage followed by five inferential parts: priors, observation, likelihood, posterior inference and checks. Each part writes checkpoints to disk, so that an interrupted run can resume without changing the statistical design.

Claim	Rule	Prior	Reads as
$H1$ supported	$\Pr(\bar{\beta} < \log 0.8 \mid D) \geq 0.95$	0.34	at least 20% attenuation at a lock
$H1$ refuted	$\Pr(\lvert\bar{\beta}\rvert < \log 1.1 \mid D) \geq 0.95$	0.14	any effect is smaller than 10%
$H2$ supported	$\Pr(\sigma_\phi < \log 1.1 \mid D) \geq 0.95$	0.30	phase moves the deficit by under 10%
$H3$ location	two-label fit wins LOO, $\theta_S, \theta_P < 0$	—	dips sit on both predicted grids
$H3a$, $H3b$	$\Pr(\lvert\kappa_F - 1\rvert < 0.1 \mid D) \geq 0.95$	0.05	that branch tracks its own parameter
$M1$ supported	$\Pr(\delta_O < 0 \mid D) \geq 0.95$, overlap term wins LOO	0.50	overlap reduces the deficit

Table 3: *Decision rules.* ***Prior*** *is the probability each rule already scored under the priors of Table 2: a posterior counts as evidence only in as much as it has moved away from that value. A claim is supported at* 0.95*, refuted at* 0.05*, and otherwise inconclusive.*

Part 0 (Setup). The notebook installs only missing dependencies, locates the repository, and creates the checkpoint directories. The design constants, including sampling rate, analysis band, context window, prediction horizon and the retrained geometries of Table 4, are imported from a shared probing library as described in Signal Generation and Testing.

The Bayesian sampler that will be used in Parts 3–5 is also configured here: the NUTS algorithm explores the posterior distribution with 4 independent chains, each discarding 2000 initial warm-up steps used to calibrate the step size and retaining the subsequent 2000 draws as posterior samples; `target_accept` $= 0.9$ controls the sampler's step-size adaptation, with higher values producing more cautious but more precise exploration.

Part 1 (Priors). Before any Chronos output is loaded, the notebook builds the design skeleton: one planned row per geometry, lock frequency, background realisation, generator and phase. It then declares the priors of Table 2, the decision thresholds of Table 3, and the prior-sensitivity ladder $\{0.25, 0.5, 1.0\}$, three scales at which the prior width of Model A will be varied in Part 5 to verify that the H1 conclusion is driven by the data rather than by the choice of scale. Prior predictive checks simulate data from the prior alone, without any Chronos output, to verify that the declared priors are centred on no effect, admit effects large enough to falsify or support the hypotheses, and make the decision thresholds reachable from both sides. The prior specification is saved as both JSON and Parquet, making later drift between the notebook and the report detectable.

Part 2 (Observation). A collection module loads, for each geometry, the retrained Chronos-Bolt weights, generates the synthetic signals, runs the predictions, and writes five tables in which each row is one observation and each column is one variable:

- `contrasts`: one row per triplet (lock frequency f_k with its two controls $f_k \pm \delta$), sharing background realisation and phase. Each row carries the forecast amplitude recovery R at the three frequencies and the contrast d. Feeds Models A and C; the phase index is retained for H2.
- `mdl_cells`: one row per probed frequency f_c, probe stage and geometry. Each row carries the prequential codelength of a linear probe separating $f_c - 1\,\text{Hz}$ from $f_c + 1\,\text{Hz}$. Feeds Model B.
- `mdl_bandtasks`: one row per band task, geometry and probe stage. Retained only as a descriptive cross-check of overall decodability; not used by any Bayesian model.
- `collapse`: one row per swept frequency and geometry. Each row carries the across-patch token dispersion $z_g(f)$. Feeds Model D1.
- `sites`: one row per detected collapse site and branch. Each row carries the fundamental f_1 and the measured spacing $\hat{\Delta}$. Feeds Model D2.

Collection is checkpointed per geometry: a completed geometry is reloaded rather than recomputed, so an interrupted run resumes from the last incomplete geometry without altering the statistical design. The generated TSMixup and KernelSynth backgrounds are archived alongside the results, so that the exact signals behind any posterior can be inspected.

Part 3 (Likelihood). The notebook defines the five statistical models (A, B, C, D1, D2) as PyMC specifications, each built from the corresponding table of Part 2. A shared sampling wrapper fits every model with the same NUTS settings declared in Part 0, so that no model receives a more or less careful exploration than another. Before fitting the real observations, a parameter-recovery test is run on Model A, the central and most complex model. Synthetic contrasts are generated from Model A itself on the real experimental structure (the same geometries, lock frequencies, backgrounds and phases that the actual data follow), fixing the population effect $\bar{\beta}$ at four known values: 0 (no attenuation), $\log 0.8$ (20%), $\log 0.5$ (50%) and $\log 0.1$ (90%). For each scenario the model is refitted on these synthetic data and the posterior of $\bar{\beta}$ is compared against the known value. If the posterior recovers it, the experimental design has enough statistical power to detect the effect or mark a genuine absence rather than insufficient design.

Part 4 (Posterior inference). Each model is fitted with NUTS or reloaded from disk. For each hypothesis the notebook reports the 95% credible interval of the estimand, the posterior probability of the decision threshold of Table 3, and the ROPE (Region Of Practical Equivalence) probability. Where the analysis defined competing formulations, leave-one-out cross-validation (LOO) selects the one that better predicts unseen data: each observation is virtually removed in turn, the model is refitted on the remainder, and its prediction of the held-out point is scored. The formulation with the higher total score is preferred. The comparisons are:

- Model A across configuration-level covariate sets, testing whether overlap or patch size mitigates the effect (M1);
- Model B with and without probe-stage and geometry corrections (H1 representational);
- Model C with and without per-phase offsets, testing whether phase affects the deficit (H2);
- Model D1 across four grid labellings — stride only, patch only, both, neither — testing which branch generates the collapse sites (H3a, H3b).

Part 5 (Checks and verdicts). The final part tests whether each posterior can be trusted before interpreting it. The checks are not uniform across models because they target different risks:

- **Convergence** (all models): three preregistered diagnostics are verified for every fit. $\hat{R} < 1.01$ measures agreement across the four independent chains; values near 1 indicate that each chain has converged to the same distribution. Bulk and tail effective sample size above 1000 counts the genuinely independent posterior draws after correcting for autocorrelation. Zero divergences requires that the sampler encountered no geometric pathology. A model that fails any criterion is not interpreted.
- **Posterior predictive** (Model A): data are generated from the fitted posterior and compared against the observed contrasts, both pooled and stratified by configuration and generator. Model A is selected because it is the most complex and hierarchical; a simpler model that passes convergence is unlikely to fail this check.
- **Prior sensitivity** (Model A): the model is refitted across the scale ladder $\{0.25, 0.5, 1.0\}$ and with a Gaussian likelihood replacing the Student-t_4. The H1 conclusion is reportable only if the decision probability and credible interval remain stable across all variants; otherwise the conclusion reflects the choice of prior, not the data.
- **LOO reading rule**: the comparisons defined in Part 4 are collected. The key quantity is the difference in expected log pointwise predictive density (elpd_diff), which measures how much better one formulation predicts unseen data than the other; a difference smaller than two standard errors ($|\text{elpd\_diff}| < 2 \cdot \text{dse}$) is reported as inconclusive rather than rounded into a decision.

The notebook finally writes a verdict table applying the three-way rule of Table 3 to every hypothesis: supported, refuted or inconclusive.

Signal Generation and Testing

The experimental evaluation uses three synthetic signal generators: a single-harmonic sinusoidal generator, a modified TSMixup generator, and a KernelSynth generator. The last two are based on methods proposed for Chronos training.[6]

Synthetic Probing Setup

Single-Harmonic Sinusoidal Generator This generator produces a simple sinusoidal signal with a single frequency component. The generated signal can be expressed mathematically as:

$$x(t) = A \cdot \sin(2\pi f t + \phi) \tag{24}$$

The continuous signal is then sampled at a specified rate to create a discrete time-series. The parameters of the generator, such as amplitude, frequency, and phase, can be adjusted to create different sinusoidal signals for testing purposes.

Light TSMixup Generator We propose a controlled probing variant of the TSMixup method described in the Chronos data augmentation procedure.[6] The original TSMixup algorithm samples subsequences from real training datasets, applies mean scaling to each sampled time series, draws mixing coefficients from a symmetric Dirichlet distribution, and returns their convex combination.

Light TSMixup preserves this stochastic mixing structure, but replaces the real dataset collection with a controlled pool of sinusoidal signals. Each base signal is defined by frequency, amplitude, phase, length, and sampling frequency. This simplification makes the generated signals easier to interpret when testing whether Chronos-Bolt exhibits frequency-dependent blind spots or phase-locking effects.

The parameters of the generator, such as the maximum number of sinusoidal components, Dirichlet concentration, target length, and sampling frequency, can be adjusted to create different synthetic signals for testing purposes. The pseudocode for the Light TSMixup generator is listed in Appendix B.

Run	**Patch** (P)	**Stride** (S)	**overlap**	**tokens**
1	8	4	0.50	511
2	8	8	0.00	256
3	16	4	0.75	509
4	16	8	0.50	255
5	16	12	0.25	170
6	16	16	0.00	128
7	24	8	0.67	254
8	24	12	0.50	169
9	24	16	0.33	127
10	24	20	0.17	102
11	24	24	0.00	85
12	32	8	0.75	253
13	32	12	0.62	169
14	32	16	0.50	127
15	32	20	0.38	101
16	32	24	0.25	85
17	32	32	0.00	64

Table 4: *The extended configurations retained for the analysis. All are retrained from scratch under the same* 100*k-step budget and from the same seed (*42*).*

KernelSynth Generator Since TSMixup is able to produce only simple composition of sinusoidal signals, we implement KernelSynth as a more complex alternative, which can generate signals with richer frequency content and temporal dynamics. By comparing the performance of Chronos-Bolt on signals generated by both methods, we can gain insights into how structural aliasing affects the model's ability to learn and forecast time-series data with varying levels of complexity.

The KernelSynth generator is also described in the Chronos data augmentation procedure.[6] It produces synthetic signals by combining different kernel functions, using randomly selected weights from a normal distribution. The selected kernels are combined using sum and product operators. Its pseudocode, kernel bank and parameters are listed in Appendix C.

Experimental Protocol

All signals are sampled at $f_s = 512\,\text{Hz}$. Pure sinusoids are swept from 2 to 250 Hz in one-hertz steps at several non-redundant starting phases, so that every tone is fully representable under the Nyquist–Shannon theorem and a loss observed at a predicted lock frequency cannot be attributed to undersampling. A single seed (42) controls all stochastic generation, ensuring full reproducibility.

The complete experimental grid is built before any data are collected: one row per combination of geometry (P, S), lock frequency, background type, and phase. For each experimental condition, 200 background draws from both the Light TSMixup and KernelSynth generators are archived alongside the probing results, so that the exact inputs behind every posterior can be reloaded for verification. Each geometry is probed independently: the retrained Chronos-Bolt weights are loaded, the synthetic signals are generated, predictions are run, and the outputs are written to the observation tables that feed the Bayesian models of Bayesian Analysis.

The geometries listed in Table 4 are chosen to allow independent variation of P and S. Series at fixed P and varying S (P=16 with $S \in \{4, 8, 12, 16\}$; P=24 with $S \in \{8, 12, 16, 20, 24\}$; P=32 with $S \in \{8, 12, 16, 20, 24, 32\}$) isolate the stride effect, while series at fixed S and varying P (S=8 with $P \in \{8, 16, 24, 32\}$; S=12 with $P \in \{16, 24, 32\}$; S=16 with $P \in \{16, 24, 32\}$) isolate the patch effect. This factorial structure is what lets Models A and D of Bayesian Analysis separate the two branches and estimate the overlap and patch-size covariates without confounding.

Behavioral and Representational Analysis

The first analysis sweeps pure sinusoids from 2 to 250 Hz in one-hertz steps, at a sampling frequency of $f_s = 512\,\text{Hz}$. For each tone, the frequency produced by Chronos is compared with the known input frequency. Pure sinusoids keep the test easy to interpret: no background spectral content can explain a loss at a predicted lock frequency. The band stops below the Nyquist limit $f_s/2 = 256\,\text{Hz}$, so every tone swept is

P	S	O	**tokens**	$S \mid P$	$P \mid S$
8	8	0.000	256	yes	yes
16	8	0.500	255	yes	**no**
16	12	0.250	170	**no**	**no**
16	16	0.000	128	yes	yes
24	24	0.000	85	yes	yes

Table 5: *The five initial geometries. The token count is the number of patch tokens at the* 2048*-sample training context. The last two columns are divisibility conditions:* $S \mid P$ *means every stride lock is also a patch null, and* $P \mid S$ *the converse. Both hold only at* $P = S$*, where the two branches are the same set of frequencies.*

Model	f_s/S [Hz]	f_s/P [Hz]	stride only	both	patch only
`p8-s8`	64	64	—	64 128 192	—
`p16-s8`	64	32	—	64 128 192	32 96 160 224
`p16-s12`	42.67	32	42.67 85.33 170.67 213.33	128	32 64 96 160 192 224
`p16-s16`	32	32	—	32 64 96 128 160 192 224	—
`p24-s24`	21.33	21.33	—	21.33 42.67 64 85.33 106.67 128 149.33 170.67 192 213.33 234.67	—

Table 6: *The locked frequencies of the five geometries of Table 5, with each frequency placed under the branch that produces it. A geometry with empty outer columns is one where $S \mid P$: the two branches coincide everywhere and no frequency is attributable to either alone.*

fully representable under the Nyquist–Shannon theorem, as Patch-Based Tokenization: a Formal Description sets out, and a loss observed here cannot be attributed to undersampling.

Table 5 summarises the patch size P, stride S, overlap $O = (P - S)/P$, and token count of each configuration. All models, including the P=S=16 baseline, are trained from scratch with the same 100k-step budget and seed; the published `chronos-bolt-tiny` checkpoint is not used. This keeps training length fixed across configurations while the token count shows the computational cost of overlap.

A locked frequency can satisfy the stride condition, the patch condition, or both. The notation $S \mid P$ means that P is an integer multiple of S. In that case every stride-lock site is also a patch-null site, so there are no stride-only sites, although patch-only sites may remain. Conversely, $P \mid S$ means that every patch-null site is also a stride-lock site, so there are no patch-only sites. If neither relation holds, both types of exclusive site are possible; if $P = S$, the two sets coincide.

Table 6 applies this rule to the sweep by sorting each candidate frequency into stride only, both, or patch only. Its main purpose is attribution: a dip at an exclusive site can be associated with one branch, whereas a dip at a shared site cannot separate their contributions.

Each configuration is swept over the same band at several non-redundant starting phases. The nominal 512-sample context is shortened, when necessary, to a whole number of strides so that the final patch is not padded; otherwise padding could create an artificial collapse. When the stride divides 512, integer-frequency tones also complete whole cycles in the window. For other strides, the context differs by fewer than S samples. Each model is loaded, measured, and released in turn.

Frequency Reconstruction

This reading is behavioural and records what the model outputs. The sinusoid is given as context, the median forecast is appended and fed back, and the rollout continues to 512 generated samples. Its dominant frequency is read as the largest magnitude of the discrete Fourier transform after the mean is removed and the zero-frequency bin skipped, then averaged over phases and plotted against the input frequency. A point on the diagonal is a tone rebuilt correctly, a flat segment is a set of distinct inputs collapsing onto one output, and the shaded band is one standard deviation over phases.

Figure 1 plots the dominant frequency of the rollout against the frequency of the input tone. The dotted diagonal is a perfect reconstruction and the dashed verticals are $\mathcal{F}_{lock}$.

The P=16, S=16 geometry is that of the published Chronos-Bolt checkpoint, but only the setting is shared: the model evaluated here is not that checkpoint; it is a variant retrained from scratch under the same

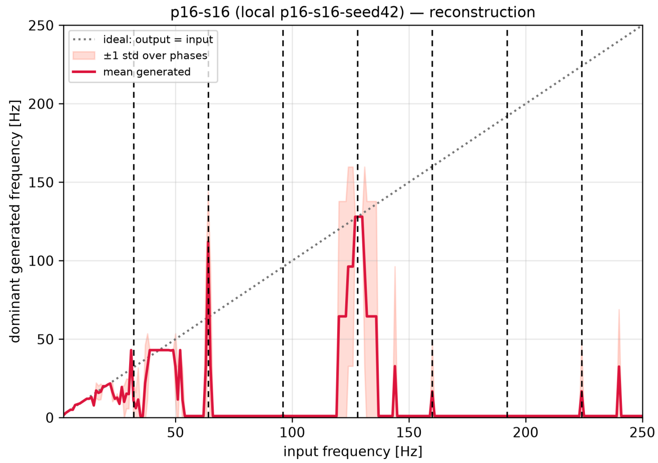

Figure 1: *Reconstruction on the $P = 16, S = 16$ geometry: dominant frequency of the rollout against the tone supplied, averaged over phases, one standard deviation shaded. The dotted diagonal is a tone rebuilt correctly; the dashed verticals are $\mathcal{F}_{lock}$.*

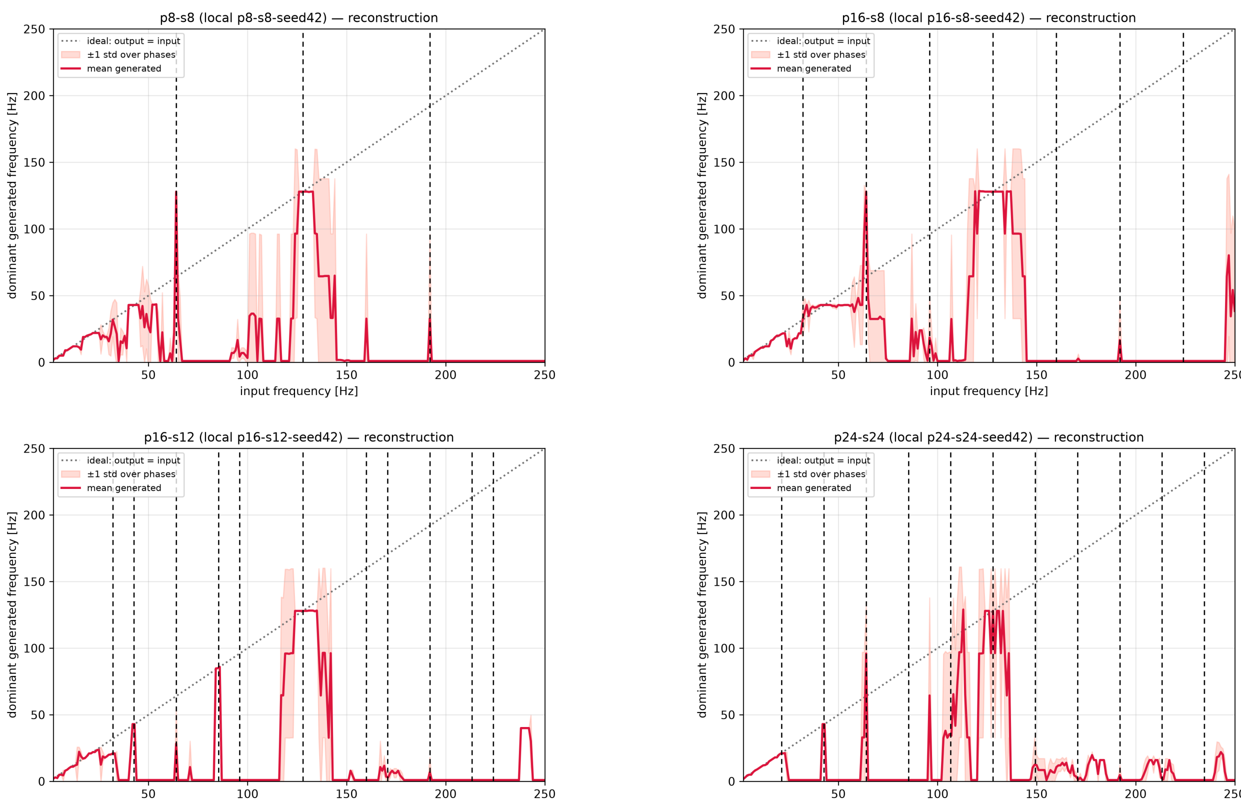

Figure 2: *Reconstruction on the other four geometries: dominant frequency of the rollout against the tone supplied, averaged over phases, one standard deviation shaded. The dotted diagonal is a tone rebuilt correctly; the dashed verticals are $\mathcal{F}_{lock}$.*

100k-step budget. Its reconstruction curve leaves the diagonal over wide stretches of the band. The figure does not licence a claim about structural aliasing: it is behavioural, and records what the model outputs rather than what its representation retains.

Figure 2 repeats the reading on the other four geometries, and the pattern is the same in each of them: every model rebuilds some frequencies correctly while others are not rebuilt at all.

Spectral Recovery

This reading is representational and records what the model still represents. The internal state is captured at the projected quantile head, and a linear probe (standardisation, then a principal component reduction, then logistic regression) is scored by cross-validation. The reading here is taken at that one stage; Model B of Bayesian Analysis repeats it at ten. Two task families are run. The seven band tasks each ask whether a tone falls in the lower or upper half of its band; their folds are grouped by frequency, so every phase of a held-out frequency is excluded from training and the probe has to generalise the boundary instead of memorising individual frequencies. An ungrouped split inflates those rows. The local task poses one question of constant difficulty at every frequency, whether $f - 1\,\mathrm{Hz}$ can be separated from $f + 1\,\mathrm{Hz}$, and holds out phases rather than frequencies; having no boundary artefact, it is the row the cross-geometry comparison uses. Cell colour is that cross-validated accuracy on a continuous scale from chance to one, so a partial loss is rendered as such rather than rounded into a bin; grey marks a frequency no task covers, which is a statement about the design and not about the model. Dashed markers on both panels are $\mathcal{F}_{lock}$ for that geometry.

Figure 3 and Figure 4 ask the same band of the representation instead of the forecast. In the local task the accuracy drops are narrow and fall close to some of the predicted lock frequencies; in the band tasks the drops instead extend over a range of frequencies around them.

These observations do not establish structural aliasing by themselves, but they do show that the models represent certain frequencies with more difficulty than others. They are the empirical ground on which the Bayesian models of Bayesian Analysis are built. The Bayesian fits share a fixed 480-sample probing context across geometries, so a stride that does not divide it can be swept but not fitted.

Limits of this design

The five geometries of Table 5 bound what can be concluded from the sweep above. Three of them have $P = S$, so their two branches coincide at every locked frequency and no observation in those geometries can be attributed to one branch. Stride-only sites exist in one geometry, P=16, S=12, and patch-only sites in two, so a claim about the stride branch rests on a single configuration and a claim about the patch branch on two. The configuration axes are equally short: the overlap ratio takes three values, and exactly one pair shares a stride while differing in patch size, which is the only place an effect of the patch can be separated from an effect of the stride. Every statement is therefore conditional on a narrow design. Attributing an effect to one branch, and testing whether either branch moves with the spacing it predicts, is beyond what these five runs can support: they can show that some frequencies are lost, not which of the two grids accounts for them. Both properties require the extended set of Table 4.

Bayesian Results

The models of Bayesian Analysis are fitted on two subsets of the grid (Table 4): a *clean* set of 15 configurations whose strides divide the probing context (CTX = 480), and a *full* set of 22 that adds seven geometries with shorter or odd strides. Posteriors are from the clean set unless stated otherwise; qualitative conclusions hold on both.

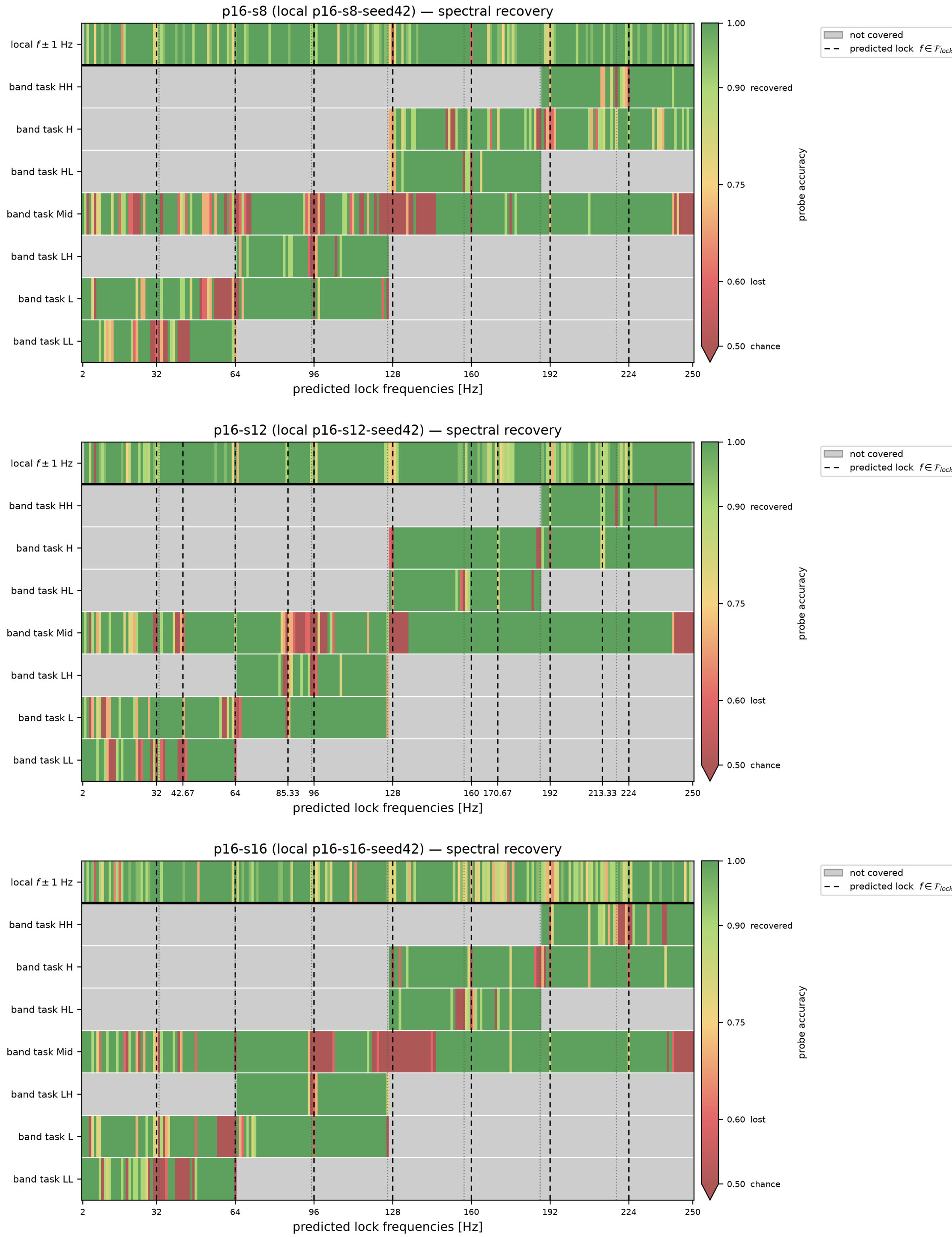

Figure 3: *Spectral recovery at $P = 16$, the stride lengthening from $S = 8$ to $S = 16$, so a feature that moves between panels belongs to the stride branch. Rows are the seven band tasks and, above the rule, the local $f \pm 1$ Hz task; colour is probe accuracy from chance to one, grey a frequency the task does not cover, dashed verticals $\mathcal{F}_{lock}$.*

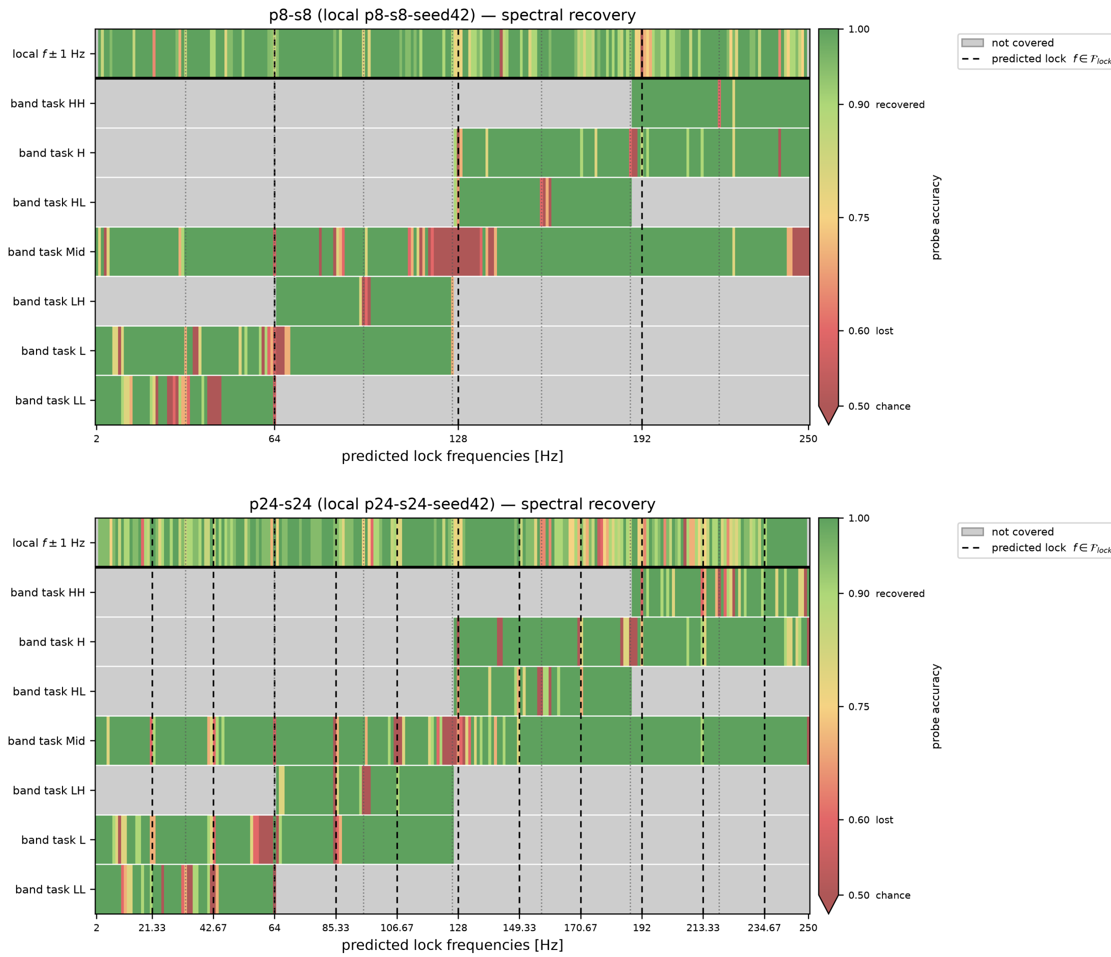

Figure 4: *Spectral recovery at the two $P = S$ geometries, where the patch length varies at zero overlap and the two branches coincide in both. Rows are the seven band tasks and, above the rule, the local $f \pm 1$ Hz task; colour is probe accuracy from chance to one, grey a frequency the task does not cover, dashed verticals $\mathcal{F}_{lock}$.*

Validation

A parameter-recovery test on Model A fixes $\bar{\beta}$ at four known values (0, $\log 0.8$, $\log 0.5$, $\log 0.1$). Every 95% credible interval covers the truth, and the posterior standard deviation shrinks from the prior width of 0.71 to 0.04–0.18; the full set passes with tighter intervals (0.04–0.11).

Models D1 and D2 pass every convergence criterion ($\hat{R} \leq 1.003$, ESS > 1000, zero divergences). Models A and C do not ($\hat{R} > 1.3$, ESS < 12), and Model B falls short on ESS (676 bulk clean, 525 full). No posterior from a non-converged model is used to *support* a claim at the 0.95 threshold.

H1: Behavioural and representational effects

Behavioural (Model A) The population effect is $e^{\bar{\beta}} = 1.08$ [0.45, 2.49] (clean) and 1.02 [0.61, 1.70] (full). The probability of at least 20% attenuation is 0.19 and 0.21; the ROPE probability is 0.13 and 0.25: the verdict is **inconclusive**. Figure 5 shows the per-configuration posteriors spanning both sides of zero with no systematic direction.

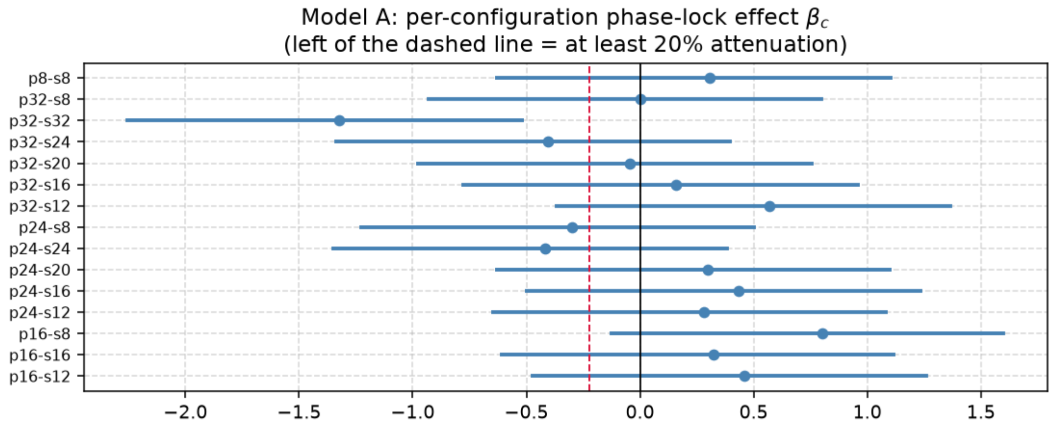

Figure 5: *Forest plot of the per-configuration phase-lock effect β_c (Model A, clean set). Each bar is the 95% credible interval; the dashed line at $\log 0.8$ marks 20% attenuation. The wide, bidirectional intervals explain the inconclusive population estimate.*

Representational (Model B) The codelength expansion at a locked frequency is $e^{\theta_{lock}} = 1.27$ [1.23, 1.31] (clean, $\Pr > \log 1.2 = 1.00$, **supported**) and 1.22 [1.19, 1.24] (full, $\Pr = 0.84$, **inconclusive**). Both agree on direction and approximate size; the difference is whether the lower tail clears the 20% mark.

The split The behavioural test finds no consistent forecast attenuation; the representational test finds a consistent internal cost. This split — *representation penalised, forecast intact* — is the central finding: the model encodes a locked frequency less efficiently yet still reconstructs it, presumably because the decoder draws on enough redundant context to compensate.

H2: Phase invariance

The posterior is $\sigma_\phi = 0.033$ [0.020, 0.067] (clean) and 0.031 [0.019, 0.067] (full), well inside the ROPE at $\log 1.1 \approx 0.095$. The ROPE probability is 0.997 in both runs, against a prior of 0.30: the verdict is **supported**. Model C has convergence issues ($\hat{R} > 1.6$), but the margin is wide enough (0.997 vs 0.95) that the conclusion survives. Phase randomisation is therefore not a viable mitigation.

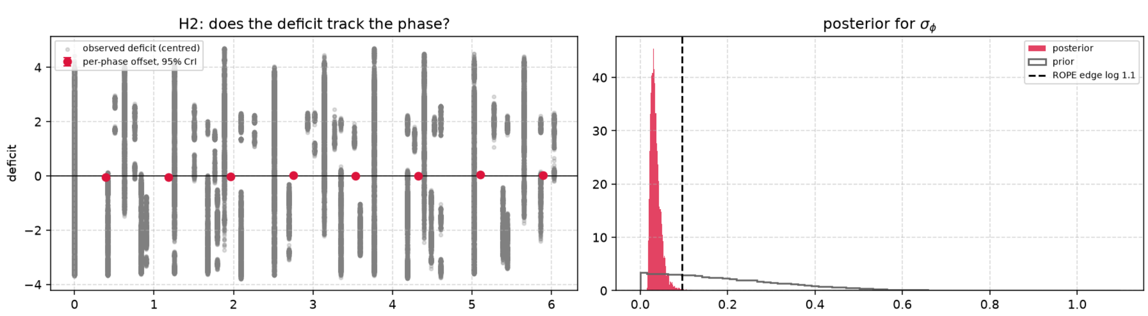

Figure 6: *Posterior for the phase term of H2. Left: observed deficit against phase ϕ with per-phase offsets u_p. Right: posterior for σ_ϕ against its prior, the dashed line marking the ROPE edge. Model C has not met convergence thresholds, so the posterior is provisional.*

H3: Token-collapse sites

All D1 models converge ($\hat{R} \leq 1.003$, ESS > 5000). The stride-only label (M_S) wins LOO for TSMixup; the two-label model (M_{SP}) wins for pure sinusoids. In every comparison $\theta_S < 0$, confirming that frequencies on the stride grid have lower token dispersion. Figure 7 shows the dips sitting on the stride grid and shifting when the stride changes.

The stride scaling slope is $\kappa_S = 1.000$ [0.992, 1.008] (clean) and 1.006 [0.990, 1.022] (full), with $\Pr(|\kappa_S - 1| < 0.1) = 1.00$ in both runs against a prior of 0.048: $H3a$ is **supported**. The patch branch ($H3b$) is **not identified**: only 9 out of 180+ rows carry a valid fundamental, because in most configurations $S \mid P$ and no observation can be attributed to the patch branch alone.

Sensitivity and concordance

Model A is refitted across the prior-scale ladder $\{0.25, 0.5, 1.0\}$ and with a Gaussian likelihood (Figure 8). The clean-set median of $\bar{\beta}$ moves from -0.05 to $+0.29$ to -0.31; the full set shows the same instability ($+0.06$ to -0.19 to $+0.17$). The 20%-attenuation probability ranges from 0.00 to 0.62 (clean) and 0.05 to 0.43 (full). The H1 behavioural conclusion is prior-dependent and cannot be reported as data-driven.

Where Bayesian and frequentist analyses address the same question, they agree: $H1$ behavioural inconclusive, $H2$ supported, $H3a$ supported. The well-converged models (D1, D2) deliver the firmest findings; the poorly converged ones (A, B, C) deliver qualified ones.

Mitigation Strategy

The mitigation pre-registered in Bayesian Analysis is increased patch overlap: shorter stride S at fixed P, raising $O = (P - S)/P$. The stride branch places its lowest in-band member at f_s/S, so halving S doubles that frequency and thins the family. The Bayesian test is Model A′ ($M1$ in Table 3), which fits the overlap-centred covariate δ_O.

The $M1$ test

The posterior of δ_O is 0.44 [-0.03, 0.81] (clean) and 0.38 [0.01, 0.72] (full). Both are *positive*: higher overlap is associated with a larger contrast, not a smaller one. The probability that $\delta_O < 0$ is 0.036 (clean) and 0.023 (full), far below the 0.95 threshold: the verdict is **refuted**.

Two interpretations are consistent. First, shortening S at fixed context multiplies the token count by P/S, changing the effective sequence length; this confound means the geometry effect and the sequence-length effect cannot be separated. Second, overlap thins only the stride branch while leaving the patch branch untouched, so the relative weight of the patch component grows and the aggregate contrast can move in the unexpected direction.

Arithmetic limits

Even setting the confound aside, the arithmetic bounds what overlap can achieve. The stride fundamental $f_{1,S} = f_s/S$ rises as S falls and can be thinned to any degree, but the patch fundamental $f_{1,P} = f_s/P$ depends on P alone: every integer multiple of f_s/P remains a potential lock site regardless of S. On the one-minute photovoltaic telemetry of the ontology domain ($f_s = 1/60\,\mathrm{Hz}$, $P = 16$), the patch harmonics have periods of 1 to 16 minutes and stay blind under any stride reduction.

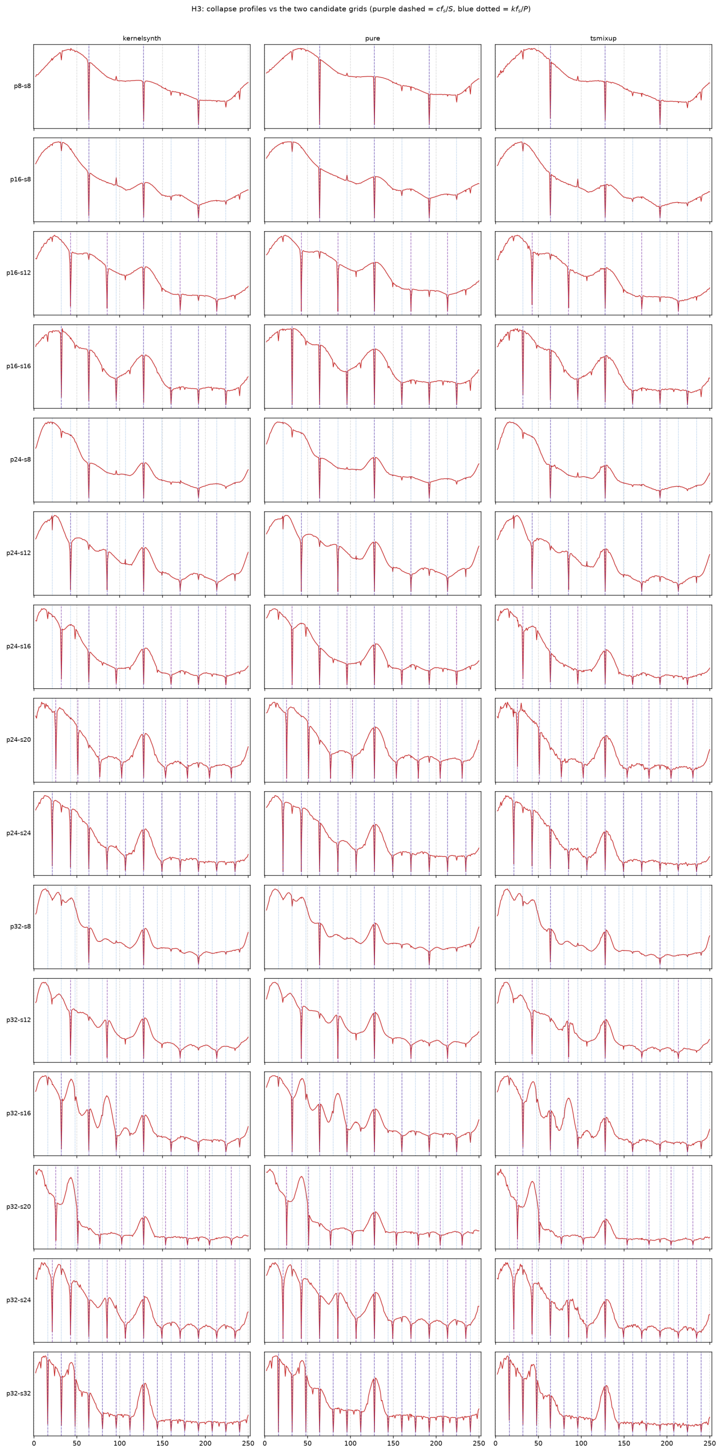

Figure 7: *Collapse profiles against the two candidate grids. Rows are configurations, columns generators; the stride grid cf_s/S is dashed and the patch grid kf_s/P dotted. Dips that follow the dashed grid as the stride changes belong to the stride branch.*

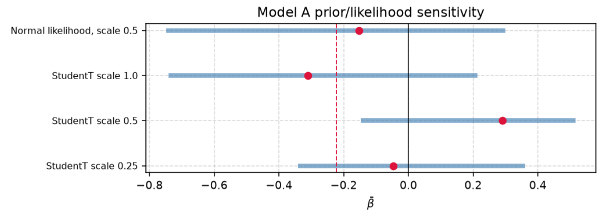

Figure 8: *Prior and likelihood sensitivity for Model A. The posterior of $\bar{\beta}$ is shown for three prior scales and for the Gaussian variant. The median shifts sign across variants.*

Alternative directions

Three untested directions could address the patch branch: increasing P to raise $f_{1,P}$ and thin the patch family at the cost of coarser temporal resolution; a learnable tokeniser (e.g. a 1-D convolution with a trained stride) that steers its frequency nulls away from signal content; or a multi-resolution front end feeding tokens from two or more (P, S) pairs whose nulls fall at different frequencies.

The firmest practical conclusion is negative: an operator who increases the overlap ratio cannot rely on that change alone, because the component of the deficit generated by the patch grid is invariant to the stride.

Agentic AI

Computer Security

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Appendix A: Ontology of the Photovoltaic Application Domain

This document defines a compact ontology for a photovoltaic system governed by one control agent. It connects environmental and electrical observations to anomaly recognition, Chronos-style patched representations, structural-aliasing evidence, and bounded power-routing decisions [6]. Routing is included only as the operational endpoint of the semantic model; the principal concern remains whether the representation preserves the information on which the agent relies.

Domain Description

The modeled system contains four energy assets: a photovoltaic array, battery storage, an aggregate internal load, and an external grid. The internal load is a passive sink representing a factory or a fleet of machines. Its demand is observed, but it has no agent, machine-level taxonomy, curtailment command, or scheduling model. The external grid is an available bidirectional boundary bus. It supplies residual demand and accepts residual production.

Exactly one `pv:ControlAgent` coordinates the four passive assets. It observes one synchronized control cycle, assigns environmental, anomaly, and observability states, selects a control mode, and emits one or more directional decisions. There is no peer-to-peer negotiation among assets.

The seven admissible transfer directions are

$$\begin{array}{rcc} \text{PV} \to \text{Load}, & \text{PV} \to \text{Battery}, & \text{PV} \to \text{Grid}, \\ \text{Battery} \to \text{Load}, & \text{Battery} \to \text{Grid}, & \text{Grid} \to \text{Load}, \\ & \text{Grid} \to \text{Battery}. & \end{array} \tag{25}$$

They satisfy the cycle-level balance

$$P_{\text{pv}} + P_{\text{bat,dis}} + P_{\text{grid,in}} = P_{\text{load}} + P_{\text{bat,ch}} + P_{\text{grid,out}}. \tag{26}$$

All terms denote non-negative magnitudes; import and export are therefore kept separate rather than encoded by an ambiguous sign.

Purpose and Scope

The ontology provides the smallest semantic layer needed to represent the physical boundary, validated observations, environmental context, three selected anomalies, and the effect of patching on model observability. It does not model individual machines, market prices, dispatch optimisation, grid faults, or a deployed real-time controller. Load demand, battery SoC, asset limits, and the optional grid-exchange reference are asserted or simulated because they are not present in the photovoltaic telemetry dataset.

The environmental states are operational generation proxies, not meteorological diagnoses. Likewise, the routing rules describe a deterministic conceptual policy rather than the central experimental contribution. Structural aliasing remains an empirically tested property of a signal-representation pair; no fixed synthetic frequency is transferred to the one-minute photovoltaic signal.

Ontology Representation

Core classes and closed vocabularies Table 7 defines the compact class interface. The four asset subclasses form a disjoint union; each `pv:PVSystem` contains exactly one asset of each type and is managed by exactly one control agent. Closed operational vocabularies are represented as named individuals rather than deeper class hierarchies.

Class	Role
`pv:PVSystem`	Aggregate system and scope of each control cycle.
`pv:EnergyAsset`	Parent of the four passive physical assets.
`pv:PhotovoltaicArray`	Intermittent renewable source measured through PV telemetry.
`pv:BatteryStorage`	Bounded storage asset that can charge, discharge, or hold.
`pv:InternalLoad`	Aggregate passive demand of a factory or machine fleet.
`pv:ExternalGrid`	Reliable bidirectional boundary bus for residual balancing.
`pv:ControlAgent`	The single decision-making entity.
`pv:SystemObservation`	Time-stamped snapshot of PV, environment, battery, load, and grid reference.
`pv:TimeSeriesSignal`	Physical signal and its temporal descriptors.
`pv:PatchedRepresentation`	A model input representation defined by patch and stride.
`pv:ObservabilityAssessment`	Evidence linking a signal-representation pair to an observability state.
`pv:GenerationContext`	Operational proxy for expected solar generation.
`pv:AnomalyState`	Detected physical or sensing abnormality.
`pv:ObservabilityState`	Semantic assessment of information retained after patching.
`pv:ControlMode`	Normal, conservative, or safe operating policy.
`pv:PowerRoute`	Closed vocabulary of the seven admissible source–destination pairs.
`pv:PowerRoutingDecision`	One directional transfer or hold decision with an optional bounded setpoint.

Table 7: *Core classes in the simplified ontology.*

The closed vocabularies are:

Generation context: ClearSky, PartialCloud, Overcast, and Nighttime.
Anomaly: InverterTrip, RampEvent, and SensorMalfunction.
Observability: Observable, PartiallyObservable, and StructurallyAliased.
Control mode: NormalMode, ConservativeMode, and SafeMode.
Power route: PVToInternalLoad, PVToBattery, and PVToExternalGrid;
BatteryToInternalLoad and BatteryToExternalGrid;
ExternalGridToInternalLoad and ExternalGridToBattery.

Formally, each vocabulary class is an OWL-style enumeration of the listed named individuals (`oneOf`). The five vocabulary classes are pairwise disjoint, and all listed values are declared globally different (`AllDifferent`). This prevents a value such as SafeMode from being silently identified with ClearSky or NormalMode under open-world semantics. Each valid time-stamped observation has exactly one generation context. Each completed observability assessment has exactly one observability state, and each routing decision records exactly one control mode. Co-issued decisions based on the same synchronized observation must record the same mode. Anomalies are not declared mutually exclusive because, for example, an abrupt inverter trip also creates a large power change. Deterministic control is obtained through the priority rules in the closed-loop section.

Object properties The semantic relations in Table 8 connect the physical boundary to the inference and control layers. Each active `pv:PowerRoutingDecision` has exactly one route value, one source, and one distinct destination. Its non-negative setpoint magnitude is optional because `Balance` may denote passive residual exchange. A cycle that simultaneously serves the load, charges the battery, and exports a residual uses three instances of the same decision class, all based on the same observation and mode.

Property	**Domain**	**Range or meaning**
`pv:hasAsset`	`pv:PVSystem`	`pv:EnergyAsset`
`pv:managedBy`	`pv:PVSystem`	exactly one `pv:ControlAgent`
`pv:hasObservation`	`pv:PVSystem`	`pv:SystemObservation`
`pv:hasSignal`	`pv:PhotovoltaicArray`	`pv:TimeSeriesSignal`
`pv:hasRepresentation`	`pv:TimeSeriesSignal`	`pv:PatchedRepresentation`
`pv:assessesRepresentation`	`pv:ObservabilityAssessment`	exactly one `pv:PatchedRepresentation`
`pv:hasGenerationContext`	`pv:SystemObservation`	zero or one `pv:GenerationContext` value
`pv:hasAnomaly`	`pv:SystemObservation`	zero or more `pv:AnomalyState` values
`pv:hasObservabilityState`	`pv:ObservabilityAssessment`	one `pv:ObservabilityState` value
`pv:selectsMode`	`pv:PowerRoutingDecision`	exactly one `pv:ControlMode` value
`pv:makesDecision`	`pv:ControlAgent`	`pv:PowerRoutingDecision`
`pv:basedOn`	`pv:PowerRoutingDecision`	observation, assessment, or inferred state
`pv:hasRoute`	`pv:PowerRoutingDecision`	zero or one `pv:PowerRoute` value
`pv:hasSourceAsset`	`pv:PowerRoutingDecision`	zero or one `pv:EnergyAsset`
`pv:hasDestinationAsset`	`pv:PowerRoutingDecision`	zero or one `pv:EnergyAsset`

Table 8: *Object-property signatures.*

`pv:hasRepresentation` is intentionally non-functional because one signal can be evaluated under several (P, S) configurations. Each assessment identifies exactly which representation it evaluates. Each decision is based on exactly one current observation and its completed assessment; its own `pv:selectsMode` value therefore scopes the mode without accumulating incompatible modes on the persistent agent. The range of `pv:basedOn` is the union of observation and inferred-assessment entities, rather than the intersection that multiple RDFS range declarations would imply.

OWL's open-world semantics do not by themselves validate that endpoint types match a named route. The operational validator therefore admits only the seven ordered pairs in Equation 25, rejects identical endpoints, and checks consistency among `pv:hasRoute`, source, and destination. A hold decision has no route or endpoints and has a zero or absent setpoint. The `pv:basedOn` relation is retained only as the minimum evidence link required to justify a selected mode; no separate trace model is introduced.

Data properties and units The synchronized observation combines the measured photovoltaic and environmental variables with asserted or simulated battery, load, and grid-reference values. The control-critical quantities are PV power P_{pv}, tilted irradiance G_{tilt}, normalized battery state of charge $s \in [0, 1]$, and aggregate load demand $P_{\mathrm{load}} \geq 0$. Each has a separate validity flag derived from presence, finiteness, freshness, and a configured physical range. Horizontal irradiance, air temperature, wind speed, and wind direction remain environmental evidence but do not independently block the controller. The signed `pv:gridExchangeReferenceW` is optional, with positive values denoting export; it is usable only when present, finite, and fresh.

Missing or stale data cannot be inferred safely from OWL's open-world assumption alone. An operational validation layer therefore compares the observation time stamp with a configurable staleness limit, validates numeric ranges, and asserts the four critical Boolean flags consumed by the ontology. This keeps data-quality validation distinct from semantic classification.

Owner	**Data properties**
`pv:SystemObservation`	`pv:pvPowerW`, `pv:powerRampRateWPerS`, `pv:loadDemandW`, optional `pv:gridExchangeReferenceW`, `pv:tiltedIrradianceWm2`, `pv:horizontalIrradianceWm2`, `pv:airTemperatureC`, `pv:windSpeedMs`, `pv:windDirectionDeg`, `pv:batterySoC`, `pv:observationTimestamp`, `pv:pvPowerValid`, `pv:tiltedIrradianceValid`, `pv:batterySoCValid`, and `pv:loadDemandValid`.
`pv:BatteryStorage`	`pv:capacityWh`, `pv:reserveSoC`, `pv:targetSoC`, `pv:maximumSoC`, `pv:maximumChargePowerW`, and `pv:maximumDischargePowerW`.
`pv:TimeSeriesSignal`	`pv:signalFrequencyHz`, `pv:samplingFrequencyHz`, `pv:periodSamples`, `pv:amplitudeW`, `pv:timeSpanSeconds`, and `pv:phaseOffsetSamples`.
`pv:PatchedRepresentation`	`pv:patchSize`, `pv:stride`, and derived `pv:overlapRatio`.
`pv:ObservabilityAssessment`	`pv:analysisStatus`, `pv:assessmentMethod`, `pv:assessmentMetric`, `pv:strideLockingRatio`, `pv:patchLockingRatio`, `pv:lockingTolerance`, `pv:effectEstimate`, `pv:posteriorLossProbability`, `pv:credibleIntervalLower`, `pv:credibleIntervalUpper`, and `pv:assessmentTimestamp`.
`pv:PowerRoutingDecision`	optional non-negative `pv:powerSetpointW`, `pv:decisionConfidence`, `pv:batteryCommand` in {Charge, Discharge, Hold}, `pv:pvCommand` in {Connect, Isolate}, and `pv:gridCommand` in {Import, Export, Balance}.

Table 9: *Quantitative property groups and units encoded by their names.*

A valid patched representation satisfies $P \geq 1$ and $1 \leq S \leq P$. Overlap is descriptive metadata:

$$\gamma = \frac{P - S}{P}. \tag{27}$$

It is not used as a standalone rule for declaring aliasing.

Inference Rules

Environmental generation context Let $g_0 < g_1 < g_2$ be site-configurable tilted-irradiance thresholds and let G_t denote the validated value of G_{tilt} at time t. The active generation context is

$$C_G(G_t) = \begin{cases} \textsf{Nighttime}, & G_t \leq g_0, \\ \textsf{Overcast}, & g_0 < G_t < g_1, \\ \textsf{PartialCloud}, & g_1 \leq G_t < g_2, \\ \textsf{ClearSky}, & G_t \geq g_2. \end{cases} \tag{28}$$

Values $g_0 = 0$, $g_1 = 150$, and $g_2 = 800\ \mathrm{W\,m^{-2}}$ are retained only as an illustrative configuration compatible with the current coursework. They are not universal meteorological boundaries. In particular, low irradiance can also be caused by solar position; the four values must therefore be read as generation-context proxies rather than certain weather diagnoses.

Anomaly detection and precedence Let $v_P(t)$, $v_G(t)$, $v_s(t)$, and $v_L(t)$ denote the validity of PV power, tilted irradiance, battery SoC, and aggregate load demand. The control-critical validity predicate is

$$V_t = v_P(t) \land v_G(t) \land v_s(t) \land v_L(t). \tag{29}$$

If V_t is false, no physical anomaly or forecast-driven route is inferred from that observation. A valid irradiance value may still support a descriptive generation context, but the operational validator asserts SensorMalfunction and the cycle proceeds directly to safe-mode selection. Within a validated cycle, absence of the three listed anomaly values is interpreted locally as “none detected”; this is a scoped closed-world policy, not a global OWL assumption. The signed ramp rate, measured in watts per second when Δt is expressed in seconds, is

$$r_t = \frac{P_t - P_{t-1}}{\Delta t}. \tag{30}$$

The anomaly rules are expressed with configurable thresholds: g_{trip} for sufficient irradiance, p_0 for near-zero power, and $r_{\max}$ for maximum admissible ramp magnitude.

$$\begin{aligned} \neg V_t &\longrightarrow \mathsf{SensorMalfunction}, && (31)\\ V_t \wedge G_t \geq g_{\text{trip}} \wedge |P_t| \leq p_0 &\longrightarrow \mathsf{InverterTrip}, && (32)\\ V_t \wedge V_{t-1} \wedge |r_t| > r_{\max} &\longrightarrow \mathsf{RampEvent}. && (33) \end{aligned}$$

No numerical value is inherited for $r_{\max}$ because it must be calibrated after invalid values and outliers have been excluded. The controller evaluates sensor validity first. A sensor malfunction therefore cannot be reinterpreted as a physical ramp or inverter trip. If a valid inverter trip also satisfies the ramp predicate, the safe response to the trip takes precedence over ramp smoothing.

Signal observability and structural aliasing For a signal frequency f, sampling frequency f_s, patch size P, and stride S, define a stride-lock ratio and a patch-periodicity ratio

$$\rho_S = \frac{fS}{f_s}, \qquad \rho_P = \frac{fP}{f_s}. \tag{34}$$

A stride phase-lock candidate L_S and a within-patch repetition candidate L_P are

$$\begin{aligned} L_S &= \left[\exists m \in \mathbb{Z}^+ : |\rho_S - m| \leq \varepsilon_S\right],\\ L_P &= \left[\exists k \in \mathbb{Z}^+ : |\rho_P - k| \leq \varepsilon_P\right],\\ C &= (L_S \vee L_P) \wedge \left(0 < f < \frac{f_s}{2}\right), \end{aligned} \tag{35}$$

where ε_S and ε_P are tied to frequency resolution and are never hardcoded frequency bands. Only L_S expresses equality of consecutive patch phases; L_P records whole-cycle repetition inside a patch and is retained as an H3 experimental factor. Their union Z is a candidate geometry class, not proof that two domain concepts are indistinguishable.

This is a division of labour, and it bounds what the ontology can conclude on its own. Phase-locking is *derivable*: L_S, L_P and Z follow by rule from `pv:signalFrequencyHz`, `pv:samplingFrequencyHz`, `pv:patchSize` and `pv:stride`, all of which the knowledge base already holds, so a reasoner can enumerate the candidate set for any configuration without observing a model. Structural aliasing is *not* derivable and is only recorded: it enters through Y below, which is posterior evidence from an experiment external to the ontology, and no rule here can produce it. An agent using this ontology therefore detects phase-locking and reports structural aliasing.

Let q denote the task-relevant information being tested, for example frequency or amplitude. Let $\Delta_{\text{loss}}(q)$ be its predeclared behavioral or representational loss relative to matched controls, δ the maximum practically negligible loss, D the experimental evidence, and $\tau > 1/2$ the posterior confidence threshold. Define confirmed information loss Y and confirmed preservation R as

$$Y = \left[\Pr(\Delta_{\text{loss}}(q) > \delta \mid D) \geq \tau\right], \qquad R = \left[\Pr(\Delta_{\text{loss}}(q) \leq \delta \mid D) \geq \tau\right]. \tag{36}$$

Because $\tau > 1/2$, Y and R cannot both hold for the same assessment. The assessment interface is designed to receive the planned offline evidence already specified in Deliverable 1: a local amplitude-recovery contrast with a heavy-tailed Student-t_4 likelihood and a Gamma regression with log link for non-negative prequential-MDL codelength. The population contrast uses the weakly informative prior $\bar{\beta} \sim \text{Student-}t_4(0, 0.5)$; the Gamma model uses $\theta_{\text{lock}} \sim \mathcal{N}(0, 0.5)$ and shape $k \sim \text{Gamma}(2, 0.1)$. The stored record identifies the tested (P, S) configuration, matched controls, method and metric, effect estimate, 95% credible interval, and posterior probability of at least 20% attenuation. These are evidence fields, not universal ontology thresholds. Until that planned inference has completed and satisfies the predeclared criterion, no instance may be asserted as StructurallyAliased.

The three observability values are assigned as follows:

$$C_O = \begin{cases} \mathsf{StructurallyAliased}, & X, \\ \mathsf{Observable}, & \neg X \wedge R, \\ \mathsf{PartiallyObservable}, & \neg X \wedge \neg R, \end{cases} \qquad X = Z \wedge Y. \tag{37}$$

Observable therefore means that positive evidence shows the tested representation preserves the task-relevant information within tolerance δ. It does not mean global observability of the physical system, nor does it merely mean that aliasing was not found. Candidate geometry may still be Observable when the experiment demonstrates preservation. PartiallyObservable is the epistemic fallback: preservation has not been certified, while structural aliasing has not been confirmed. It includes a candidate with inconclusive evidence, an unfinished test, a credible interval crossing the decision threshold, conflicting metrics, or no preservation evidence. The term means *partially trusted*, not that a fixed fraction of samples or sensors is visible.

H1 compares each candidate frequency with matched nearby non-candidate controls and predicts higher MDL codelength or lower amplitude recovery; absent or non-localized loss refutes H1. H2 tests whether the loss at a candidate frequency remains approximately stable across sampled phase offsets; systematic phase dependence refutes that hypothesis. H3 tests whether candidate degradation sites move proportionally to $1/S$ or $1/P$ when the corresponding parameter varies. These hypotheses contribute controlled evidence to D, but remain falsifiable experimental factors rather than ontology subclasses or automatic action rules. Sensor validity is modeled separately as an anomaly and never relabels a model representation as structurally aliased.

When an experimental generator provides a phase ϕ in radians, the corresponding temporal offset is converted before storage:

$$o_{\text{samples}} = \frac{\phi f_s}{2\pi f}. \tag{38}$$

Neither ϕ nor o_{samples} is compared directly with stride to infer an anomaly.

The semantic propagation path is explicit:

$$\begin{aligned} \texttt{pv:TimeSeriesSignal} &\longrightarrow \texttt{pv:PatchedRepresentation} \\ &\longrightarrow \texttt{pv:ObservabilityAssessment} \\ &\longrightarrow \texttt{pv:ControlMode} \longrightarrow \texttt{pv:PowerRoutingDecision}. \end{aligned} \tag{39}$$

Aliasing can therefore reduce confidence in forecast-derived production, ramp expectations, forecast-driven anomaly recognition, and routing. It does not alter the directly observed values of G_{tilt}, P_{pv}, load demand, or SoC; those values are governed by the separate validity path.

Closed-Loop Control

Mode selection The agent evaluates each synchronized control cycle using the strict priority

$$\mathsf{SafeMode} \succ \mathsf{ConservativeMode} \succ \text{RampOverride} \succ \text{NormalRouting}. \tag{40}$$

Let N_{safe} mean that SensorMalfunction or InverterTrip is present. The selected mode is

$$M_t = \begin{cases} \mathsf{SafeMode}, & N_{\text{safe}} \vee (C_O = \mathsf{StructurallyAliased}), \\ \mathsf{ConservativeMode}, & C_O = \mathsf{PartiallyObservable}, \\ \mathsf{NormalMode}, & \text{otherwise}. \end{cases} \tag{41}$$

In SafeMode, the battery command is Hold and the grid balances the internal load. The PV command is Isolate only for an inverter trip; for sensor malfunction or confirmed structural aliasing it remains connected, but forecast- and ramp-driven battery decisions are suspended. When load demand itself is invalid, Balance carries no invented numerical setpoint. In ConservativeMode, the same seven route types remain available with a configurable reduced battery-power cap, normal target/reserve bounds, and low decision confidence. A concurrent ramp cannot activate either extended ramp bound; this implements the precedence in Equation 40.

Routing policy The ordinary policy serves the internal load before charging the battery or exporting energy. Let

$$P_{\text{sur}} = (P_{\text{pv}} - P_{\text{load}})^{+}, \qquad P_{\text{def}} = (P_{\text{load}} - P_{\text{pv}})^{+}. \tag{42}$$

Here *daylight* abbreviates {ClearSky, PartialCloud, Overcast}. A cycle issues a route only for a strictly positive transfer and may issue several co-mode decisions. All setpoints are clipped to SoC, capacity, and charge/discharge limits.

Figure 9 shows the priority scan and Table 10 states its boundary cases. The conservative branch rejoins the ordinary load-first router under its reduced cap. The ramp branch is available only in NormalMode.

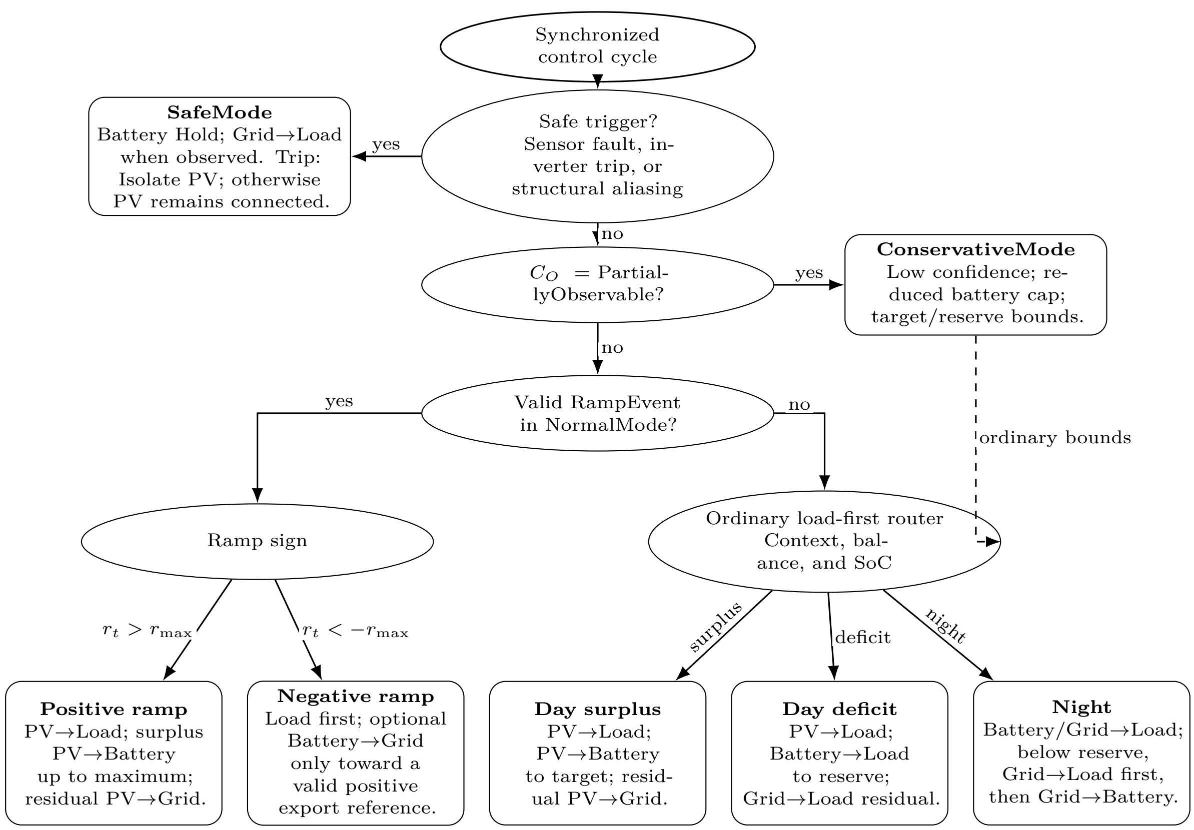

Figure 9: *Load-first routing decision tree. Elliptical bubbles are ordered gates; rounded bubbles are terminal policies. Safe and conservative decisions precede the normal-mode ramp override, while the ordinary router covers daylight surplus, daylight deficit, and nighttime operation.*

Condition	**Ordered route set**	**Constraint**
Daylight surplus	PV→Load; PV→Battery; PV→Grid	Serve demand, charge to target, then export the residual.
Daylight deficit	PV→Load; Battery→Load; Grid→Load	Discharge only to reserve; the grid covers the remaining deficit.
Night, $s > s_{\text{reserve}}$	Battery→Load; Grid→Load	Battery serves demand to reserve; the grid supplies the residual.
Night, $s \leq s_{\text{reserve}}$	Grid→Load; if $s < s_{\text{reserve}}$, Grid→Battery	Supply the load first; charge only below reserve and stop at reserve.
Positive ramp with surplus	PV→Load; PV→Battery; PV→Grid	NormalMode may charge beyond target, never beyond maximum SoC.
Negative ramp with valid positive reference	load-serving routes; Battery→Grid	Export only after serving the load, within residual discharge capacity and above reserve.
Negative ramp without usable reference	ordinary surplus/deficit routes	Missing, stale, non-finite, zero, or negative reference disables export catch-up.
ConservativeMode	ordinary load-first routes	Reduced battery cap; target/reserve bounds; no ramp extensions.
SafeMode	Grid→Load when observed	Battery Hold; grid Balance; isolate PV only for inverter trip.

Table 10: *Routing matrix for the seven admissible transfer directions.*

For a negative ramp, Battery→Grid is admissible only after the internal load has been fully served, the export reference is present, fresh, finite, and positive, measured export is below that reference, the battery remains above reserve, and discharge capacity remains after load support. If any condition fails, the ordinary load-first route applies. At night, Grid→Battery begins only after Grid→Load has been satisfied and stops when reserve is restored. These rules preserve Equation 26 without turning the routing policy into a separate optimisation problem.

Appendix B: Light TSMixup Pseudocode

Algorithm 1 Light TSMixup: Controlled Sinusoidal Time Series Mixup

Require: Sinusoidal pool $\mathcal{P} = \{p_1, \ldots, p_{N_p}\}$, where each p_n is defined by frequency f_n, amplitude A_n, phase ϕ_n, source length T_n, and sampling frequency f_s; maximum components $K = 10$; symmetric Dirichlet concentration $\alpha = 1.5$; minimum and maximum augmented lengths ($l_{\min} = 128, l_{\max} = 2048$).

Ensure: An augmented time series.

```
k ~ U{1, K}                                        ▷ number of sinusoidal components to mix
l ~ U{l_min, l_max}                                ▷ length of the augmented time series
for i ← 1 to k do
    n ~ U{1, N_p}                                  ▷ sample a sinusoidal source
    x^(i)_{1:l} ← SampleSegment(p_n, l, f_s)       ▷ generate or crop a length-l segment
    x̃^(i)_{1:l} ← x^(i)_{1:l} / ((1/l) Σ_{j=1}^{l} |x^(i)_j|)   ▷ apply mean scaling
end for
[λ_1, ..., λ_k] ~ Dir([α_1 = α, ..., α_k = α])     ▷ sample mixing weights
return Σ_{i=1}^{k} λ_i x̃^(i)_{1:l}                 ▷ take a weighted combination of scaled signals
```

In mathematical notation:

1. $k \sim U\{1, K\}$ ▷ number of sinusoidal components to mix
2. $l \sim U\{l_{\min}, l_{\max}\}$ ▷ length of the augmented time series
3. **for** $i \leftarrow 1$ to k **do**
4. $\quad n \sim U\{1, N_p\}$ ▷ sample a sinusoidal source
5. $\quad x_{1:l}^{(i)} \leftarrow \mathrm{SampleSegment}(p_n, l, f_s)$ ▷ generate or crop a length-l segment
6. $\quad \tilde{x}_{1:l}^{(i)} \leftarrow x_{1:l}^{(i)} \Big/ \left(\frac{1}{l} \sum_{j=1}^{l} \left| x_j^{(i)} \right| \right)$ ▷ apply mean scaling
7. **end for**
8. $[\lambda_1, \ldots, \lambda_k] \sim \mathrm{Dir}([\alpha_1 = \alpha, \ldots, \alpha_k = \alpha])$ ▷ sample mixing weights
9. **return** $\sum_{i=1}^{k} \lambda_i \tilde{x}_{1:l}^{(i)}$ ▷ take a weighted combination of scaled signals

Appendix C: KernelSynth Pseudocode and Kernel Bank

Algorithm 2 KernelSynth: Synthetic Data Generation using Gaussian Processes

Require: Kernel bank $\mathcal{K}$ listed in Table 11; maximum kernels per time series $J = 5$; generated length $l_{\text{syn}} = 1024$; optional numerical jitter $\epsilon = 0$.

Ensure: A synthetic time series $x_{1:l_{\text{syn}}}$.

$j \sim U\{1, J\}$ ▷ sample the number of kernels
$\{\kappa_1(t,t'), \ldots, \kappa_j(t,t')\} \overset{\text{i.i.d.}}{\sim} \mathcal{K}$ ▷ sample kernels with replacement
$t \leftarrow \text{linspace}(0, 1, l_{\text{syn}})$
$\kappa^\star(t,t') \leftarrow \kappa_1(t,t')$ ▷ initialize the composite covariance
for $i \leftarrow 2$ to j **do**
  $\star \sim U\{+, \times\}$ ▷ sample a binary composition operator
  $\kappa^\star(t,t') \leftarrow \kappa^\star(t,t') \star \kappa_i(t,t')$ ▷ compose kernels
end for
if $\epsilon > 0$ **then**
  $\kappa^\star(t,t') \leftarrow \kappa^\star(t,t') + \epsilon I$
end if
$x_{1:l_{\text{syn}}} \sim \mathcal{GP}(0, \kappa^\star(t,t'))$ ▷ sample from the GP prior
return $x_{1:l_{\text{syn}}}$

Table 11: *Kernel bank $\mathcal{K}$ used by KernelSynth.*

Kernel	Formula	Hyperparameters
Constant	$\kappa_{\text{Const}}(x,x') = C$	$C = 1$
White Noise	$\kappa_{\text{White}}(x,x') = \sigma_n \cdot \mathbf{1}_{x=x'}$	$\sigma_n \in \{0.1, 1\}$
Linear	$\kappa_{\text{Lin}}(x,x') = \sigma_0^2 + x \cdot x'$	$\sigma_0 \in \{0, 1, 10\}$
RBF	$\kappa_{\text{RBF}}(x,x') = \exp\left(-\frac{\lVert x-x'\rVert^2}{2l^2}\right)$	$l \in \{0.1, 1, 10\}$
Rational Quadratic	$\kappa_{\text{RQ}}(x,x') = \left(1 + \frac{\lVert x-x'\rVert^2}{2\alpha}\right)^{-\alpha}$	$\alpha \in \{0.1, 1, 10\}$, $l = 1$
Periodic	$\kappa_{\text{Per}}(x,x') = \exp\left(-2\sin^2\left(\pi\frac{\lVert x-x'\rVert}{p/l_{\text{syn}}}\right)\right)$	$p \in \{24, 48, 96, 168, 336, 672,$ $7, 14, 30, 60, 365, 730,$ $4, 26, 52, 4, 6, 12, 4, 40, 10\}$